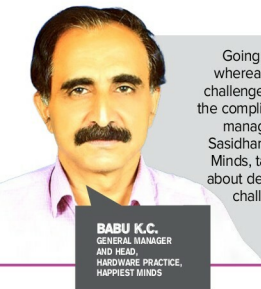


Disruptive technologies:

Arriving At A SPECIFICATION DOCUMENT... Is The BIGGEST CHALLENGE



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Going with the norm treads a relatively easy path, whereas going against the flow brings its own set of challenges. If all that you do is try to disrupt what exists, the complications are that much more. Babu K.C., general manager and head, hardware practice, and Divya Sasidharan, engineering manager, both from Happiest Minds, talk to Dilin Anand and Priya Ravindran of EFY about dealing with disruptive technologies and the challenges that crop up along the way

Q. Disruptive technologies, as the name suggests, disrupt what is already there. How challenging is it to go about creating a project specification or design flow in this domain?

A. Given the product and system specifications, arriving at a specification document that is both agreeable to the customer and feasible from a design point of view is the biggest challenge when it comes to disruptive technology. With no base to work on, coming up with an optimum architecture is also a task.

At every stage, there might be problems regarding compatibility or how the system responds to change, and it is a constant trial-and-error process. It is all about being open-minded and ready to work with, or try out new methods.

Q. What is the most disruptive technology at the moment?

A. That would be the Internet of Things (IoT). There is a tendency to connect and manage almost all gadgets or equipment remotely through the Internet. We are getting several requests for proposals for developing IoT products. Almost all new designs call for Internet connectivity by default, and many companies are looking at adding IoT features to their legacy products.

Q. How would you go about choosing a processor for a design?

A. Factors like performance and interface requirements take prominence. Speed and power requirements, cost, availability of components and support for the chip from

the vendor are also parameters we need to take into account.

Q. What are the biggest technological challenges you face while working on these projects, and how do you solve these?

A. One of the biggest challenges we face is the space constraint on the printed circuit board (PCB). Customers generally desire the boards to be as small as possible with maximum features in it.

If you add other dimensions such as low cost, we have to rule out using high-density interconnects (HDIs); although density demands having these fitting components into a limited space without using HDIs or compromising on performance, signal integrity or thermal aspects is often quite challenging.